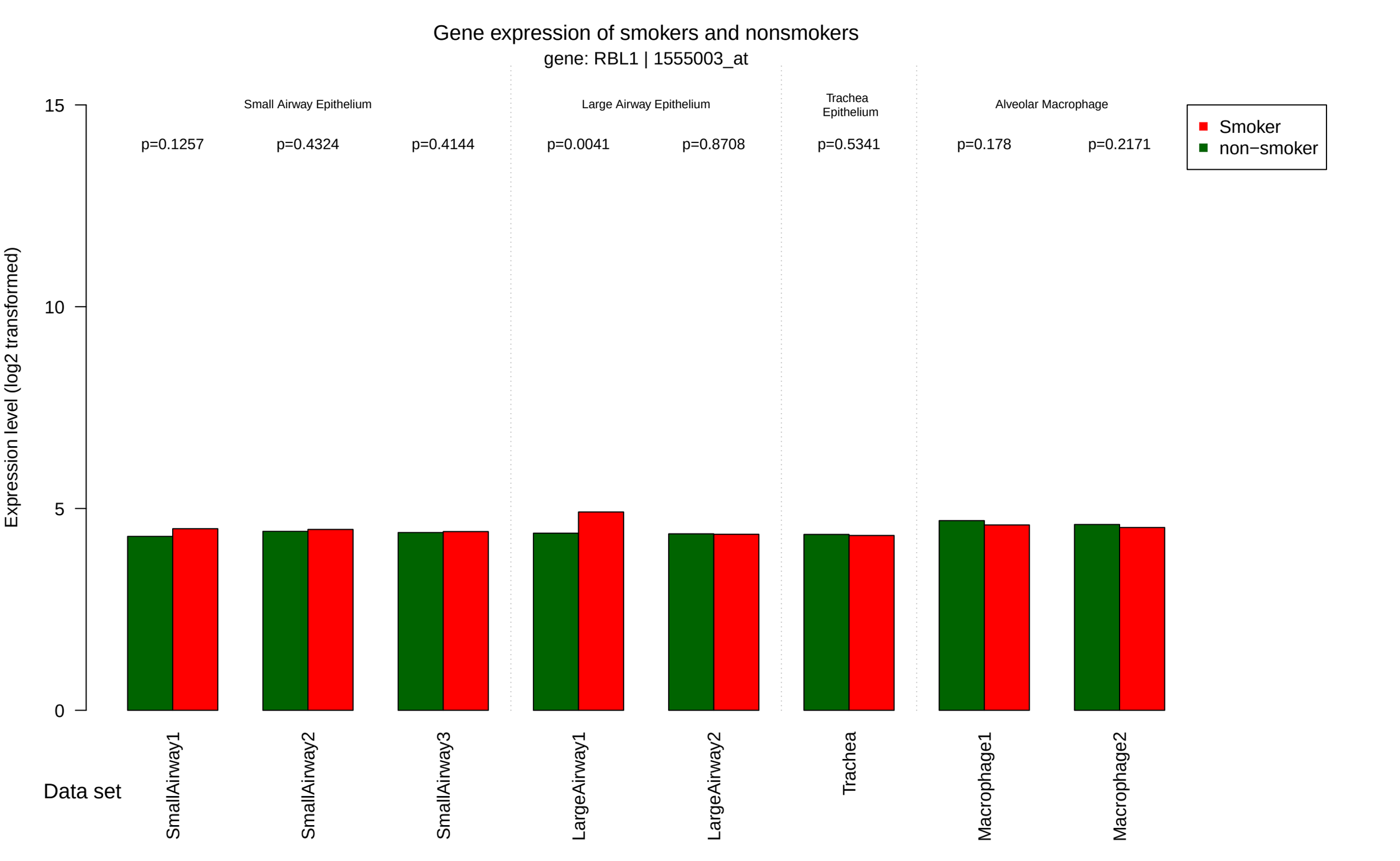
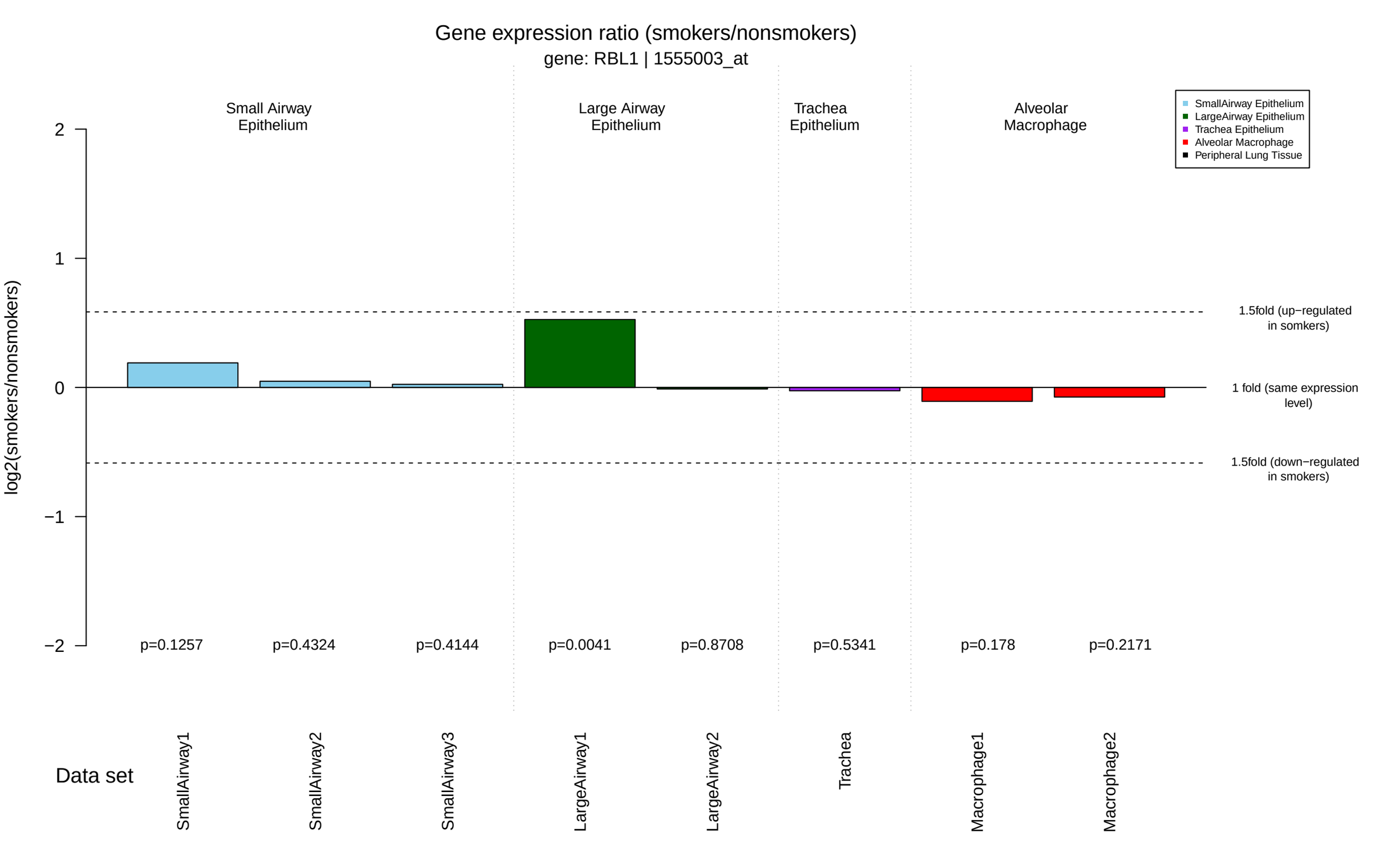
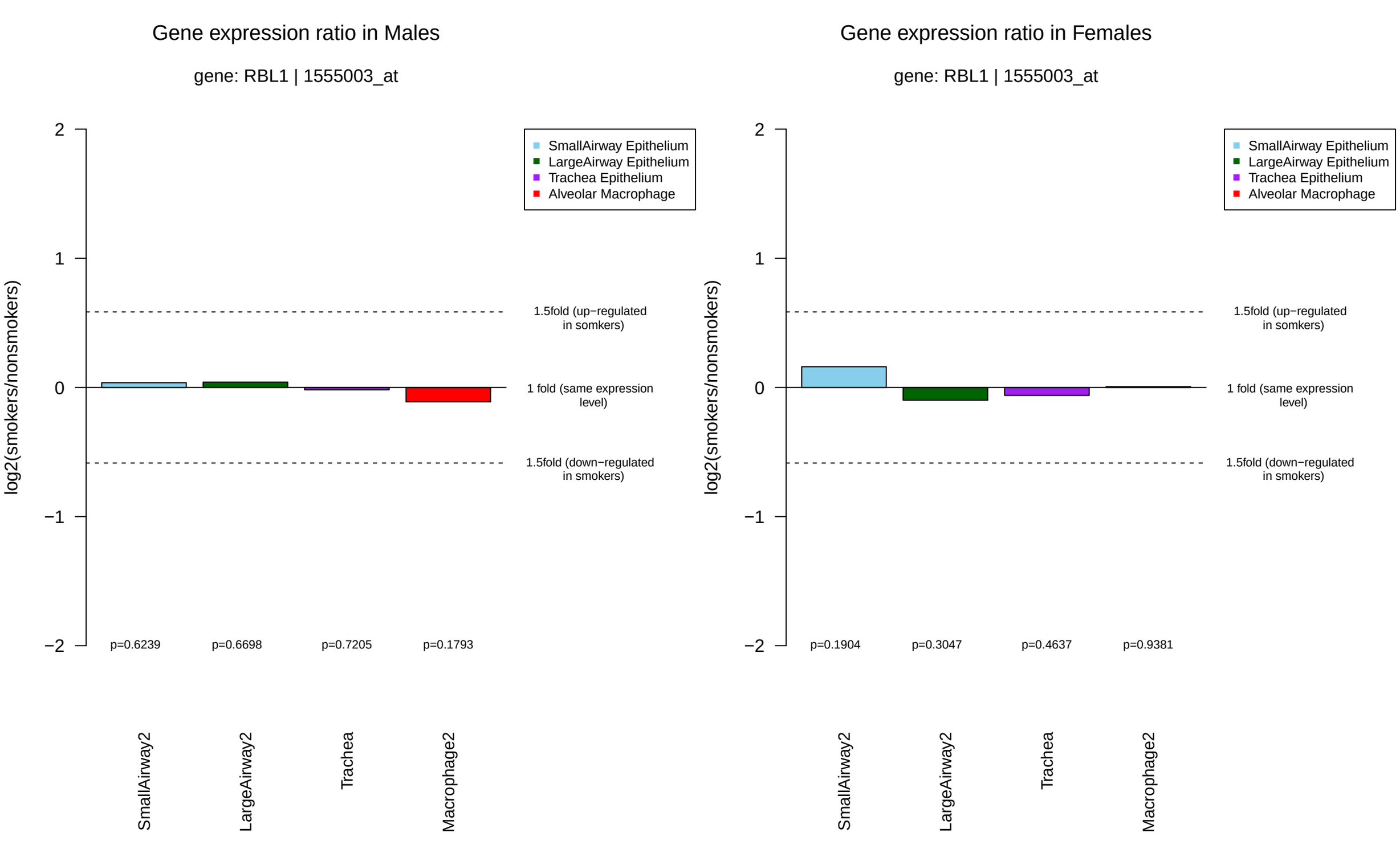
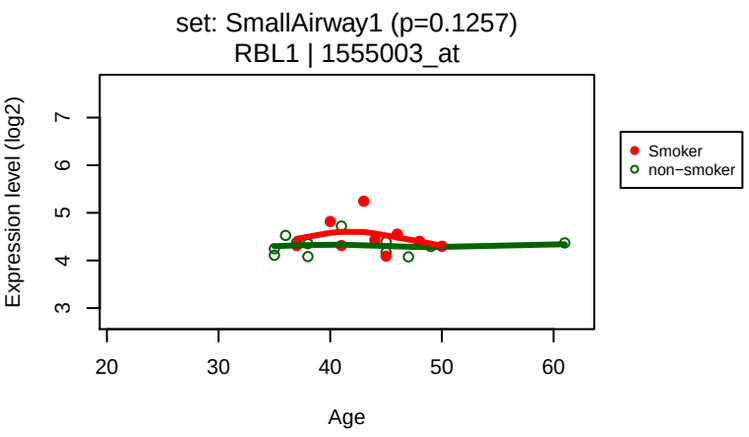
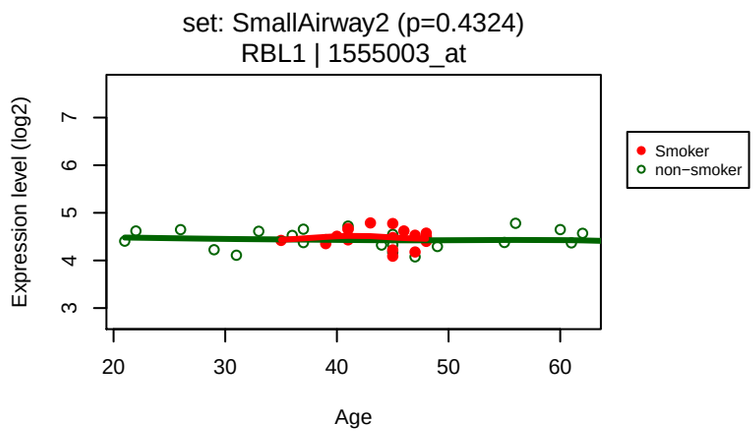
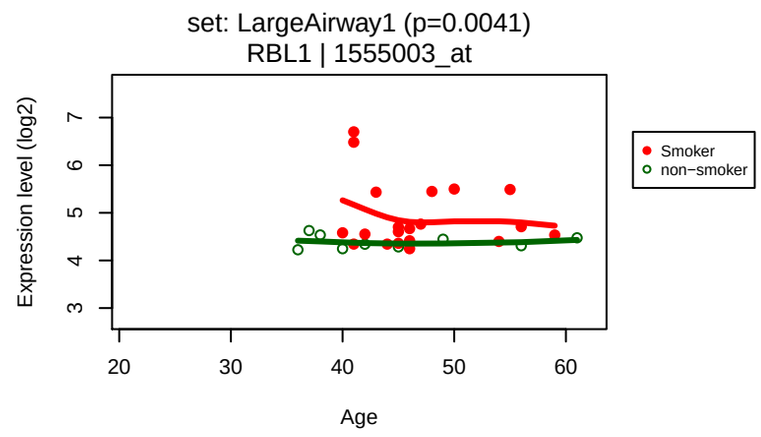

Gene expression of smokers and nonsmokers
gene: RBL1 | 1555003_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1257
p=0.4324
p=0.4144
p=0.0041
p=0.8708
p=0.5341
p=0.178
p=0.2171
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

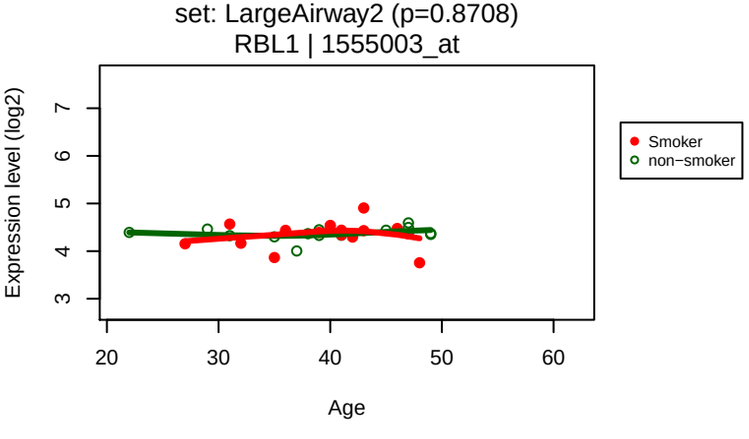

Gene expression ratio (smokers/nonsmokers)
gene: RBL1 | 1555003_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1257
p=0.4324
p=0.4144
p=0.0041
p=0.8708
p=0.5341
p=0.178
p=0.2171
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

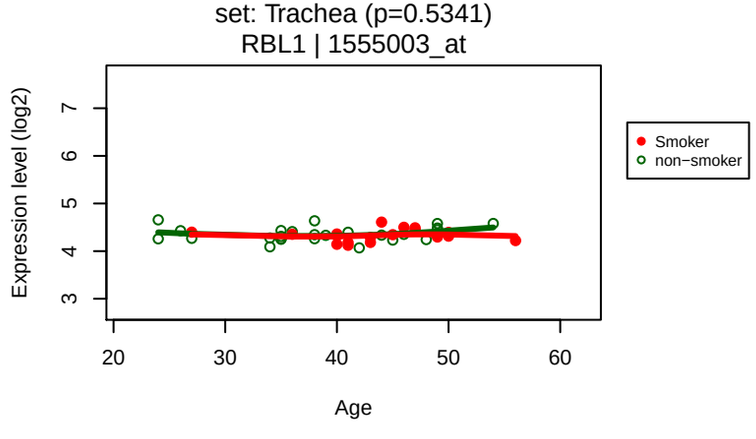

Gene expression ratio in Males
gene: RBL1 | 1555003_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6239
p=0.6698
p=0.7205
p=0.1793
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RBL1 | 1555003_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1904
p=0.3047
p=0.4637
p=0.9381
SmallAirway2
LargeAirway2
Trachea
Macrophage2

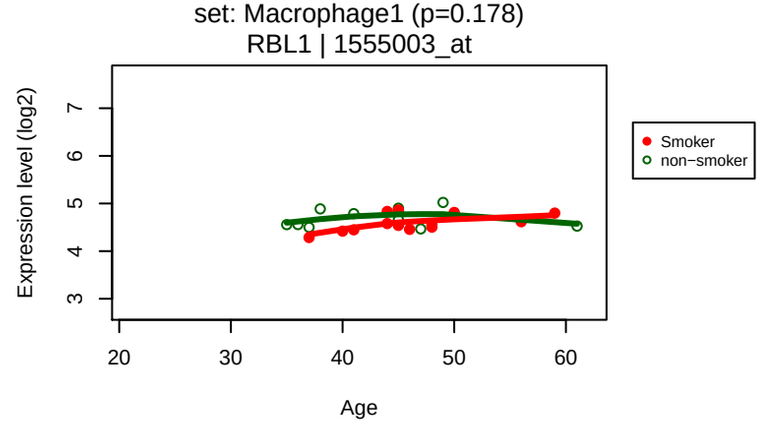

set: SmallAirway1 (p=0.1257)
RBL1 | 1555003_at
Expression level (log2)
Age
Smoker
non−smoker

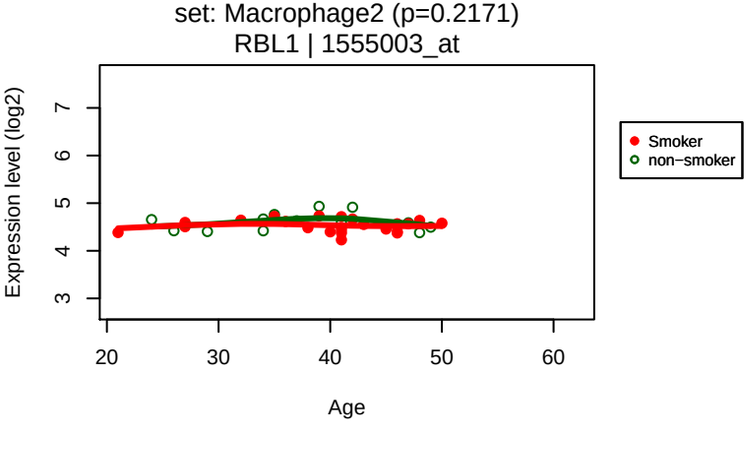

set: SmallAirway2 (p=0.4324)
RBL1 | 1555003_at
Expression level (log2)
Age
Smoker
non−smoker

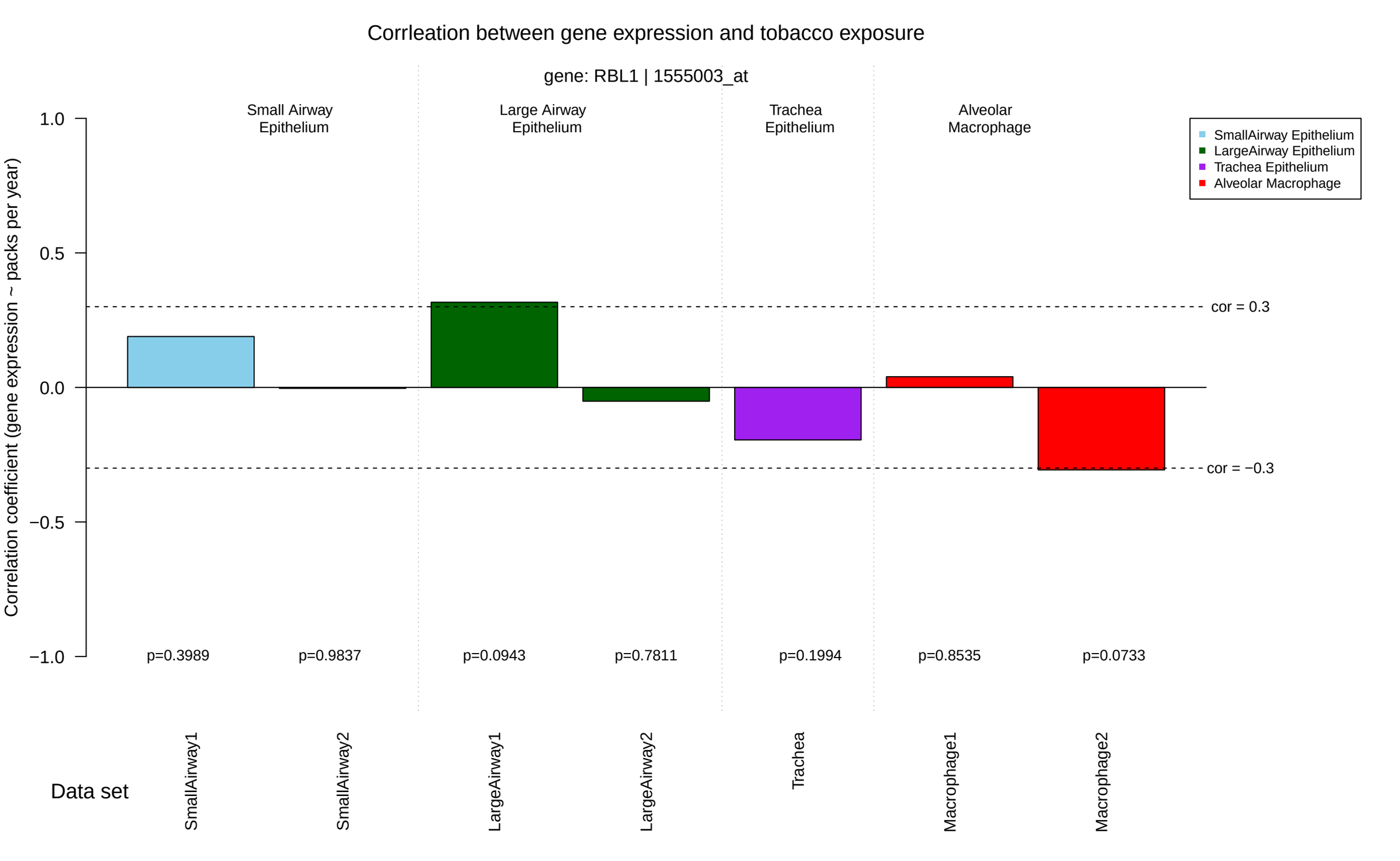

set: LargeAirway1 (p=0.0041)
RBL1 | 1555003_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8708)
RBL1 | 1555003_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5341)
RBL1 | 1555003_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.178)
RBL1 | 1555003_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2171)
RBL1 | 1555003_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RBL1 | 1555003_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3989
p=0.9837
p=0.0943
p=0.7811
p=0.1994
p=0.8535
p=0.0733
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2